

PhD Dissertation Research Proposal

Proposed Title: Integrated Cyber-Biotechnology for Precision Neural Therapeutics: A Closed-Loop CRISPR-Coupled Neuroprosthetic System with Post-Quantum Genomic Adjudication and Municipal Health Data Resilience

Candidate Location: Baltimore, MD **Proposed Supervising Department:** Biomedical Engineering, Biotechnology, Health Informatics, or Cybersecurity (cross-disciplinary) **Date:** June 2026

1. Abstract

This dissertation proposes the design, validation, and ethical/regulatory integration of a first-in-kind closed-loop neuroprosthetic platform that combines CRISPR-based neural editing with post-quantum cryptographic security to treat executive dysfunction arising from neurogenetic disorders. The system uses polygenic risk scores (PRS) to personalize epigenetic modulation and wraps all genomic and clinical data in an immutable, legally admissible evidence trail. To ensure the viability of such a system within municipal health infrastructure, the research also develops a companion post-quantum resilient municipal network architecture (the Baltimore Secure Backbone System, BSBS) designed to protect sensitive neurogenomic data from harvest-now-decrypt-later threats. The work thus spans two interconnected domains: precision neurotherapeutics and health data cybersecurity, unified by the common thread of trustworthy, verifiable, and quantum-safe biomedical computation.

2. Research Questions & Objectives

Overarching Research Question:

How can we design, validate, and safely deploy a closed-loop neuroprosthetic system that uses CRISPR-mediated epigenomic feedback for executive dysfunction, while

simultaneously securing its data flows against current and quantum-era threats and rendering them admissible in disability adjudication?

Specific Objectives:

1. CRISPR-Neural Integration Framework

- Map the complete landscape of CRISPR-Cas variants suitable for post-mitotic neurons (Cas9, base editors, prime editing, Cas13).
- Design a conceptual “CRISPRa/i adapter” for the CINP prosthetic that responds to real-time polygenic risk score fluctuations.
- Validate the theoretical safety and efficacy of such a system using in silico modeling and literature-derived case studies (SCN2A, C9orf72, COMT, DRD4).

2. Genomic Adjudication & Evidentiary Pipeline

- Develop a GPRS-Secure++ framework that translates polygenic risk evidence into standardized SSA/ADA disability criteria (Listing 12.11).
- Build the High-Security Genomic Data Generator with federated intelligence (PubMed, PACER, ScholarPK) and a SHA3-256 Semantic Evidence Trace ledger to create litigation-grade evidence objects.
- Formalize the evidence chain so that each neuroprosthetic intervention is wrapped in a cryptographically verifiable audit log.

3. Post-Quantum Municipal Health Infrastructure (BSBS)

- Model the 2019 Baltimore ransomware attack as a case study in municipal vulnerability.
- Design a zero-trust, unified network architecture (BSBS) that enforces hardware-rooted trusted execution environments.
- Develop a cryptographic migration roadmap for all city-held health and genomic data to NIST PQC standards (CRYSTALS-Kyber, CRYSTALS-Dilithium).
- Demonstrate through simulation that BSBS would prevent harvest-now-decrypt-later and ransomware attacks on neurogenomic data.

4. Ethical, Regulatory, and Policy Synthesis

- Evaluate the ethical landscape of somatic neural editing for disability, ensuring compliance with federal and international guidelines.

- Propose a governance framework for “evidentiary neuroprosthetics” under ADA, HIPAA, and the Federal Rules of Evidence (Daubert standard).
- Draft model legislation/executive order for a municipality to adopt PQC mandates.

3. Original Contribution to Knowledge

This dissertation will make the following original contributions:

- First closed-loop neuroprosthetic architecture that integrates CRISPRa/i for on-demand epigenomic tuning, governed by patient-specific polygenic risk models.
- First litigation-grade genomic evidence pipeline that binds biomedical, legal, and scholarly sources into a quantum-resistant, auditable data structure suitable for disability adjudication.
- First comprehensive municipal blueprint for securing neurogenomic and health data against quantum adversaries, extending beyond IT patching to systemic cryptographic resilience.
- A new ethical-regulatory framework for “biocybernetic” devices that act both as therapeutic tools and as sources of legal evidence.

4. Methodological Framework

The research will be structured as a multi-phase, mixed-methods project:

Phase 1: Literature and Technology Synthesis (Year 1)

- Systematic reviews of CRISPR neural editing, zero-trust architectures, and PQC standards.
- Gap analysis: No existing platform combines these.
- Formal specification of the CINP-CRISPR hybrid and BSBS requirements.

Phase 2: Component Design and In Silico Validation (Year 2)

- In silico modeling of CRISPR gRNA targeting for COMT/DRD4 using bioinformatics tools (Benchling, CRISPOR).

- Software development of the High-Security Genomic Data Generator microservices (Python, FastAPI) with PQC key management (liboqs).
- Network simulation of BSBS zero-trust model and PQC migration using open-source network simulators (GNS3, Mininet) with integrated quantum-resistant TLS.

Phase 3: Evidentiary Integration and Regulatory Analysis (Year 3)

- Build the federated evidence fusion engine that connects real PACER dockets, PubMed abstracts, and ScholarPK metrics.
- Formal evaluation of the evidentiary robustness of PRS-based claims using the Daubert standard.
- Draft the ethical and policy chapters, including stakeholder interviews (disability advocates, municipal IT leaders).

Phase 4: Dissertation and Defense (Year 4)

- Synthesis of all components into a single, cohesive dissertation.
- Preparation of a prototype demonstration (containerized microservices) for defense.
- Publication of key results in peer-reviewed journals (e.g., Nature Biomedical Engineering, IEEE Transactions on Information Forensics and Security, Journal of Law and the Biosciences).

5. Timeline (Gantt-Style Summary)

Activity	Year 1	Year 2	Year 3	Year 4
Literature review & spec	■ ■ ■ ■			
CRISPR in silico design		■ ■ ■ ■		
Genomic Generator & SET ledger		■ ■ ■ ■		
BSBS simulation & PQC plan		■ ■ ■ ■		
Evidence fusion engine			■ ■ ■ ■	
Regulatory/ethical analysis			■ ■ ■ ■	
Dissertation writing & defense				■ ■ ■ ■

6. Prior Work & Portfolio Basis

This PhD research is grounded in an extensive body of prior technical, policy, and certification work that will be directly integrated:

- **CompTIA A+/Network+/Security+, Healthcare IT Technician** – Foundational IT and health IT competency.
- **Unified Cognitive Infrastructure System Patent** – Provisional filing for the neuroprosthetic-adjudication platform.
- **High-Security Genomic Data Generator** – Already designed and partially implemented; serves as the core data infrastructure.
- **Baltimore Secure Backbone System Policy Whitepaper** – Provides the municipal cybersecurity case study and architectural skeleton.
- **Published/Preprint Research** – Privacy-preserving GWAS (Ostrak), PQC-secured genomics (Fujiwara), blockchain/TEE eHealth (Wang), and pathway PRS tools (Choi et al.) all inform the technical chapters.

The PhD will thus extend and formalize this prior work into a coherent, original contribution rather than starting from scratch—making it ideal for a portfolio-based or publication-based doctoral program, if the university allows it.

7. Potential Host Institutions & Fit

This single-track PhD would thrive in a program that values interdisciplinary research:

- **Johns Hopkins University (Baltimore):** Department of Biomedical Engineering, or the cross-divisional program in Cybersecurity & Health.
- **University of Maryland Baltimore County (UMBC):** Human-Centered Computing or Bioengineering tracks with strong cybersecurity focus.
- **George Mason University:** Bioengineering and Cyber Security Engineering dual-degree possibilities.
- Any university offering a PhD by Prior Publication could accept the core portfolio with a critical synthesis dissertation.

8. Conclusion

This single-track PhD proposal bridges two of the most urgent frontiers in 21st-century health technology: precision neurotherapeutics and post-quantum data security. By uniting CRISPR, closed-loop neuroprosthetics, polygenic risk adjudication, and municipal cryptographic resilience, the research will deliver both a novel therapeutic platform and the infrastructure to trust it. The candidate's existing certifications, patent, software, and policy papers provide a strong, validated foundation; the doctoral program will transform these into a rigorous, original, and lasting contribution to science, law, and public health.